Package ‘GTAPViz’

March 2, 2025

Title Visualization of GTAP Model Results from HAR and SL4 Files

Version 0.0.0.9000

Author Pattawee Puangchit <ppuangch@purdue.edu>

Description GTAPViz is an extension package of HARplus, designed to provide seamless data visualization for GTAP model results from `.HAR` and `.SL4` files. This package focuses on user-friendly visualization tools, enabling GTAP users to analyze and compare simulation results efficiently without requiring deep coding knowledge. With built-in functions for pivoting, mapping, unit assignments, and graphical representation, HARplusViz enhances economic model interpretation by generating structured, publication-ready plots. It is ideal for GTAP researchers, policymakers, and analysts looking for intuitive data exploration and visual insights from GEMPACK outputs.

License MIT + file LICENSE

Encoding UTF-8

Roxygen list(markdown = TRUE)

RoxygenNote 7.3.2

BugReports https://github.com/bodysbobb/GTAPViz/issues

URL https://bodysbobb.github.io/GTAPViz/

Imports HARplus,
stats,
ggplot2,
dplyr,
tidyr,
openxlsx,
colorspace,
grDevices,
rlang,
scales,
purrr,
readxl,
writexl,
utils,
methods

Remotes github::Bodysbobb/HARplus

Suggests knitr,
rmarkdown,
devtools

VignetteBuilder knitr

Exports add_unit_col,
comparison_plot,
detail_plot,
macro_plot,
gtap_macros_data,
add_mapping_info

ImportFrom HARplus get_data_by_var load_sl4x
stats setNames
ggplot2 ggplot aes aes_string geom_bar geom_text geom_vline geom_col geom_hline facet_wrap
theme_minimal theme element_text element_rect element_blank unit margin labs
scale_fill_manual scale_y_continuous position_dodge expansion
dplyr filter mutate case_when
colorspace hex2RGB hex polarLUV sRGB
grDevices col2rgb rgb
rlang sym
scales squish

Depends R (>= 3.5)

Contents

`add_mapping_info` *Add Mapping Information to GTAP Data*

Description

Adds description and unit information to GTAP data based on specified mapping.

Usage

```
add_mapping_info(
  data_list,
  external_map = NULL,
  mapping = "GTAPv7",
  description_info = TRUE,
  unit_info = TRUE
)
```

Arguments

data_list — Data structure containing GTAP variables

external_map — Optional data frame with mapping information

mapping — Mapping mode: "GTAPv7" (default), "No", "Yes", or "Mix"

description_info — Logical. If TRUE, adds description information

unit_info — Logical. If TRUE, adds unit information

Value

Data structure with added mapping information, preserving original structure

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Load example data
har_data <- HARplus::load_harx(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPViz"))

# Corrected example with closing quotation
har_data <- add_mapping_info(har_data, mapping = "GTAPv7")

## End(Not run)
```

---

comparison_plot — *Generate Comparison Plots*

Description

Creates bar plots comparing values across categories with flexible panel configuration.

Usage

```
comparison_plot(
  data,
  plot_var = NULL,
  x_axis_from,
  split_by,
  title_prefix = "",
  title_suffix = "",
  compare_by_experiment = FALSE,
  separate_figure = FALSE,
  invert_panel = FALSE,
  description_as_title = FALSE,
  output_dir = NULL,
  panel_rows = NULL,
```

```
  panel_cols = NULL,
  color_tone = NULL,
  legend_position = "none",
  width = NULL,
  height = NULL
)
```

Arguments

data	Data frame containing "Variable", "Value", "Experiment", "Unit" columns plus x_axis_from and split_by variables.
plot_var	Optional. Character vector or data frame with "Variable" column to specify which variables to plot.
x_axis_from	Column name to use for x-axis values (or panels if compare_by_experiment = TRUE).
split_by	Column name for separating plots (or panels if invert_panel = TRUE).
title_prefix	Optional text to prepend to plot titles.
title_suffix	Optional text to append to plot titles.
compare_by_experiment	If TRUE, Experiment is plotted on x-axis, x_axis_from values as panels.
separate_figure	If TRUE, creates individual figures for each panel value.
invert_panel	If TRUE, uses split_by for panels and Experiment for separation.
description_as_title	If TRUE, uses "Description" column for titles instead of "Variable".
output_dir	Optional path to save plots. If NULL, plots are only returned.
panel_rows	Optional number of panel rows. Auto-calculated if NULL.
panel_cols	Optional number of panel columns. Auto-calculated if NULL.
color_tone	Optional base color for the plot. NULL uses default ggplot2 colors.
legend_position	Position of legend: "none" (default), "bottom", "top", "left", "right".
width	Optional width in inches. Auto-calculated if NULL.
height	Optional height in inches. Auto-calculated if NULL.

Value

A list of ggplot objects or a single ggplot object depending on settings

Author(s)

Pattawee Puangchit

See Also

detail_plot, macro_plot, , macro_plot, stack_plot

Examples

```
## Not run:
# Generate comparison plot
comparison_plot(
  data = sl4.plot.data$REG,
  x_axis_from = "Region",
  plot_var = sl4plot,
  title_prefix = "Impact on",
  output_dir = output.folder,
  compare_by_experiment = FALSE,
  description_as_title = TRUE,
  separate_figure = FALSE,
  color_tone = "grey",
  width = 20,
  height = 12,
  legend_position = "bottom"
)

## End(Not run)
```

`convert_units` *Convert Multiple Units in Nested Data Structures*

Description

Convert Multiple Units in Nested Data Structures

Usage

```
convert_units(
  data,
  change_unit_from,
  change_unit_to,
  adjustment,
  value_col = "Value",
  unit_col = "Unit"
)
```

Arguments

`data`	A data structure (list, data.frame, or nested combination)
`change_unit_from`	Character vector of units to change (case-insensitive)
`change_unit_to`	Character vector of new units (same length as change_unit_from)
`adjustment`	Character vector of operations or numeric vector (same length as change_unit_from)
`value_col`	Column name containing values to adjust (default: "Value")
`unit_col`	Column name containing unit information (default: "Unit")

Value

Data structure with same format as input but with adjusted values and units

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Load example data
har_data <- HARplus::load_harx(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPViz"))

har_data <- convert_units(har_data,
change_unit_from = c("million USD"),
change_unit_to = c("billion USD"),
adjustment = c("/1000"))

## End(Not run)
```

`detail_plot` *Generate Detailed Plot*

Description

Creates a detailed plot reporting all regions or sectors depending on the variable selection. Uses a separate dataframe to specify variables to plot and their titles.

Usage

```
detail_plot(
  data,
  plot_var = NULL,
  y_axis_from,
  split_by,
  title_prefix = "",
  title_suffix = "",
  description_as_title = TRUE,
  panel_rows = NULL,
  panel_cols = NULL,
  positive_color = "#2E8B57",
  negative_color = "#CD5C5C",
  output_dir = NULL,
  top_impact = NULL,
  width = NULL,
  height = NULL,
  separate_figure = FALSE,
  invert_panel = FALSE,
  legend_position = "none",
  width_dodge = 0.5,
  width_bar = 0.4
)
```

Arguments

`data`	A data frame containing the full dataset.
`plot_var`	A data frame with "Variable" and "PlotTitle" columns specifying which variables to plot and their titles. If NULL, all variables in the data will be plotted with default titles.
`y_axis_from`	Character vector. Column names to use for the y-axis.
`split_by`	Character. Column used to separate different plots.
`title_prefix`	Optional character string to prepend to the title.
`title_suffix`	Optional character string to append to the title.
`description_as_title`	Logical. If TRUE, uses the "Description" column from plot_var as the plot title.
`panel_rows`	Optional numeric. Number of panel rows. If left blank, it will be automatically adjusted.
`panel_cols`	Optional numeric. Number of panel columns. If left blank, it will be automatically adjusted.
`positive_color`	Character. Color for positive values.
`negative_color`	Character. Color for negative values.
`output_dir`	Optional character. Directory to save the output plot.
`top_impact`	Optional numeric. Number of top impact factors to include. If left blank, all factors will be included.
`width`	Optional numeric. Width of the output plot. If left blank, it will be automatically adjusted.
`height`	Optional numeric. Height of the output plot. If left blank, it will be automatically adjusted.
`separate_figure`	Logical. If TRUE, creates separate figures for each combination of variable and category.
`invert_panel`	Logical. If TRUE, swaps the roles of Experiment and split_by in faceting and figure separation.
`legend_position`	Character. Position of the legend: "none" (default), "bottom", "top", "left", or "right".
`width_dodge`	Numeric. Dodge width for bar position adjustment. Default is 0.5.
`width_bar`	Numeric. Width of bars in the plot. Default is 0.4.

Value

A `ggplot` object or NULL if no plots could be generated.

Author(s)

Pattawee Puangchit

gtap_macros_data *Extract and Aggregate Scalar Macroeconomic Variables*

Description

Extracts scalar macroeconomic variables from multiple SL4 datasets and aggregates them into a structured data frame.

Usage

```
gtap_macros_data(
  input_dir = NULL,
  experiment_names = NULL,
  output_dir = NULL,
  output_formats = NULL,
  subtotal_level = FALSE,
  select_var = NULL
)
```

Arguments

`input_dir`	Character. Directory containing SL4 files.
`experiment_names`	Character vector. List of experiment names corresponding to SL4 files.
`output_dir`	Character (optional). Directory to save exported data.
`output_formats`	Character vector (optional). List of output formats (e.g., "csv", "xlsx").
`subtotal_level`	Logical. Whether to include subtotal levels in the processed data.
`select_var`	Character vector (optional). List of specific variable names to filter from the final result. If NULL, all variables are returned.

Value

A sorted dataframe containing processed GTAP macro data.

Author(s)

Pattawee Puangchit

See Also

add_mapping_info, process_gtap_data

Examples

```
## Not run:
# Method 1: Extract all variables
input_dir <- paste0(input.folder)
experiment <- c("EXP1", "EXP2") # File name (.sl4)
Macros <- gtap_macros_data(input_dir, experiment_names = experiment,
subtotal_level = FALSE)
```

```
# Method 2: Filter specific variables
Macros <- gtap_macros_data(input_dir, experiment_names = experiment,
                          select_var = c("qgdp", "pop", "gdpexp"))

## End(Not run)
```

---

macro_plot *Generate Macro Plot with Flexible Title Options*

Description

Creates figures of macro variables from the Macros header in the SL4. Uses a separate dataframe to specify variables to plot with options for using either variable names, PlotTitle, or Description columns for plot titles.

Usage

```
macro_plot(
  data,
  plot_var = NULL,
  title_prefix = "",
  title_suffix = "",
  compare_by_experiment = FALSE,
  description_as_title = FALSE,
  output_dir = NULL,
  panel_rows = NULL,
  panel_cols = NULL,
  color_tone = NULL,
  separate_figure = FALSE,
  width = NULL,
  height = NULL,
  legend_position = "none"
)
```

Arguments

`data` A data frame containing the full dataset with macroeconomic variables.

`plot_var` A data frame with "Variable" column and optionally "PlotTitle" or "Description" columns specifying which variables to plot and their titles. If NULL, all variables in the data will be plotted with default titles.

`title_prefix` Optional character string to prepend to the title.

`title_suffix` Optional character string to append to the title.

`compare_by_experiment`
Logical. If `TRUE`, compares experiments within x-axis categories.

`description_as_title`
Logical. If TRUE, uses the "Description" column for plot titles instead of "PlotTitle". Default is FALSE.

`output_dir` Optional character. Directory to save the output plot.

`panel_rows` Optional numeric. Number of panel rows. If left blank, it will be automatically adjusted.

panel_cols
: Optional numeric. Number of panel columns. If left blank, it will be automatically adjusted.

color_tone
: Optional character. Base color for the plot. If left blank, a default color will be used.

separate_figure
: Logical. If FALSE, combines multiple variables into one plot.

width
: Optional numeric. Width of the output plot. If left blank, it will be automatically adjusted.

height
: Optional numeric. Height of the output plot. If left blank, it will be automatically adjusted.

legend_position
: Position of the legend: "none", "bottom", "top", "left", or "right"

Value

A ggplot object or a list of plots if separate_figure = TRUE.

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
macro_plot(Macros,
           plot_var = macro.map,
           compare_by_experiment = FALSE,
           color_tone = "grey",
           output_dir = output.folder,
           description_as_title = FALSE,
           panel_rows = 2,
           width = 45,
           separate_figure = FALSE,
           legend_position = "bottom")

## End(Not run)
```

---

process_gtap_data *Process GTAP Data Automation with Flexible Output Options*

Description

Processes GTAP data from SL4 and HAR files with options for exporting and preparing plot-ready data.

Usage

```
process_gtap_data(
  experiment,
  mapping_info = "GTAPv7",
  project_dir = NULL,
  input_dir = NULL,
  output_dir = NULL,
  output_formats = NULL,
  plot_data = FALSE,
  region_select = NULL,
  sector_select = NULL,
  subtotal = FALSE,
  sl4_list_name = "sl4.plot.data",
  har_list_name = "har.plot.data",
  sl4_structure_name = "sl4.structure",
  har_structure_name = "har.structure",
  GTAPMacro_name = "GTAPMacro",
  sl4var = NULL,
  harvar = NULL,
  sl4map = NULL,
  harmap = NULL
)
```

Arguments

`experiment`	Character vector. Case names to process.
`mapping_info`	Character. Mapping mode: "GTAPv7" (default), "Yes", "No", or "Mix".
`project_dir`	Character. Path to the project folder with "in" and "out" subfolders.
`input_dir`	Character. Path to the input folder. Overrides project_dir/in if specified.
`output_dir`	Character. Path to the output folder. Overrides project_dir/out if specified.
`output_formats`	Character vector or list. Exports data in these formats (valid: "csv", "stata", "rds", "txt").
`plot_data`	Logical. If TRUE, prepares data for plotting and assigns to variables.
`region_select`	Optional vector. Specifies regions to filter the data.
`sector_select`	Optional vector. Specifies sectors to filter the data.
`subtotal`	Logical. If TRUE, includes subtotal data. Default is FALSE.
`sl4_list_name`	Character. Variable name for SL4 plotting data if generating plot data. Default is "sl4.plot.data".
`har_list_name`	Character. Variable name for HAR plotting data if generating plot data. Default is "har.plot.data".
`sl4_structure_name`	Character. Variable name for SL4 structure if generating plot data. Default is "sl4.structure".
`har_structure_name`	Character. Variable name for HAR structure if generating plot data. Default is "har.structure".
`GTAPMacro_name`	Character. Variable name for GTAP macro data if generating plot data. Default is "GTAPMacro".

sl4var	Data frame, NULL, or FALSE. Variables to extract from SL4 files. Set to NULL to extract all variables, or FALSE to skip SL4 processing.
harvar	Data frame, NULL, or FALSE. Variables to extract from HAR files. Set to NULL to extract all variables, or FALSE to skip HAR processing.
sl4map	Data frame or NULL. Mapping information for SL4 variables (with "Variable", "Description", and "Unit" columns).
harmap	Data frame or NULL. Mapping information for HAR variables (with "Variable", "Description", and "Unit" columns).

Details

This function provides a complete automation workflow for processing GTAP model outputs, with flexible output options and optional filtering for plot data.

The key parameters `sl4var` and `harvar` each accept three different input types:

- **Data frame**: Contains variable mappings with required "Variable" column. When provided, only specified variables will be extracted.
- **NULL**: Extracts all available variables from the respective file type.
- **FALSE**: Completely skips processing of that file type, allowing the function to focus only on the other file type.

The `mapping_info` parameter controls how descriptions and units are assigned:

- **GTAPv7**: Uses standard GTAPv7 definitions (default)
- **Yes**: Uses only the supplied descriptions and units from sl4map/harmap
- **No**: Does not add any descriptions or units
- **Mix**: Prioritizes supplied descriptions and units, falling back to GTAPv7 for any missing values

Value

Returns the processed data invisibly, which will not be printed to the console.

Author(s)

Pattawee Puangchit

See Also

add_mapping_info, gtap_macros_data

Examples

```
## Not run:
# Extract data with region and experiment filters
process_gtap_data(
  sl4var = NULL,
  harvar = NULL,
  sl4map = sl4map,
  harmap = harmap,
  region_select = selected_regions,
  sector_select = NULL,
  subtotal = FALSE,
```

```
  experiment = experiment,
  mapping_info = info.mode,
  project_dir = project.folder,
  plot_data = TRUE,
  output_formats = list(
    "csv" = "No",
    "stata" = "No",
    "rds" = "No",
    "txt" = "No"))

## End(Not run)
```

---

rename_GTAP_bilateral *Rename GTAP Bilateral Trade Columns*

Description

Renames bilateral trade columns in GTAP data output

Usage

```
rename_GTAP_bilateral(data)
```

Arguments

data	Data structure containing GTAP bilateral trade data

Value

Data structure with renamed bilateral trade columns

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Load Sample Data
sl4_data <- HARplus::load_sl4x(system.file("extdata/in", "EXP1-WEL.sl4",
                                          package = "GTAPViz"))
sl4_data <- rename_GTAP_bilateral(df)

## End(Not run)
```

`rename_value` *Rename Values in Data Structures*

Description

Replaces specified values in a column across nested data structures

Usage

```
rename_value(data, column_name = NULL, mapping.file)
```

Arguments

`data`	Data structure containing data to rename
`column_name`	Column name to modify. If NULL, extracted from mapping file
`mapping.file`	Data frame with OldName and NewName columns for renaming

Value

A modified data frame or list of data frames with specified values replaced.

Author(s)

Pattawee Puangchit

Examples

```
## Not run:
# Example mapping file
wefare.decomp.rename <- data.frame(
  OldName = c("alloc_A1", "ENDWB1", "tech_C1"),
  NewName = c("Allocation", "Endowment", "Technology"),
  ColumnName = "Variable",
  stringsAsFactors = FALSE
)

# Load example data
har_data <- HARplus::load_harx(system.file("extdata/in", "EXP1.sl4",
                                           package = "GTAPViz"))

# Apply renaming
modified_data <- rename_value(har_data, mapping.file = wefare.decomp.rename)

## End(Not run)
```

report_table *Generate a Detailed Report Table*

Description

Transforms one or more datasets into a wide-format table based on specified grouping columns and pivot column(s), with optional subtotal filtering and export.

Usage

```
report_table(
  data_list,
  pivot_col,
  total_column = FALSE,
  export_table = FALSE,
  separate_file = FALSE,
  output_dir = NULL,
  sheet_names = NULL,
  include_units = FALSE,
  component_exclude = NULL,
  group_by = NULL,
  rename_cols = NULL,
  var_name_by_description = TRUE,
  add_var_info = FALSE,
  decimal = 2,
  unit_select = NULL,
  separate_sheet_by = NULL,
  subtotal_level = FALSE,
  repeat_label = FALSE,
  workbook_name = "detail_results",
  add_group_line = FALSE
)
```

Arguments

data_list	A named list of data frames.
pivot_col	A named list indicating which column is pivoted into wide format for each dataset.
total_column	Logical. Adds a "Total" column summing numeric columns if TRUE.
export_table	Logical. Exports the tables to Excel if TRUE.
separate_file	Logical. If TRUE, each data frame is saved in a separate Excel file.
output_dir	Character. Directory where Excel files are written if export_table=TRUE.
sheet_names	Optional named list for custom sheet names.
include_units	Logical. If TRUE, includes the "Unit" column in grouping.
component_exclude	Optional character vector of pivoted values to exclude.
group_by	A list or character vector of grouping columns, or NULL.
rename_cols	A named list for renaming columns.

var_name_by_description
: Logical. If TRUE, replaces Variable names with Description or vice versa.

add_var_info
: Logical. If TRUE, appends the other part in parentheses.

decimal
: Numeric. Decimal places for rounding.

unit_select
: Optional. Filters rows to this unit.

separate_sheet_by
: Optional column name to split sheets.

subtotal_level
: Logical. If TRUE, includes all Subtotal values; otherwise keeps only Subtotal="TOTAL".

repeat_label
: Logical. If TRUE, repeats the first group column in exports.

workbook_name
: Character. Name of the Excel workbook (no extension).

add_group_line
: Logical. If TRUE, draws a thin line after each group.

Value

A named list of data frames, each transformed into wide format, optionally exported to Excel.

Author(s)

Pattawee Puangchit

---

stack_plot *Generate Stacked Bar Plot with Auto-Dimensions*

Description

Creates stacked bar plots with automatic total calculations and flexible grouping. Uses a separate dataframe to specify variables to plot and their titles. Auto-adjusts dimensions and panel layout if not specified.

Usage

```
stack_plot(
  data,
  plot_var = NULL,
  x_axis_from,
  stack_value_from,
  title_prefix = "",
  title_suffix = "",
  y_axis_title = NULL,
  compare_by_experiment = FALSE,
  separate_figure = FALSE,
  unstack_plot = FALSE,
  output_dir = NULL,
  panel_rows = NULL,
  panel_cols = NULL,
  color_tone = NULL,
  show_total = TRUE,
  description_as_title = TRUE,
```

```
  legend_position = "bottom",
  y_limit = NULL,
  width = NULL,
  height = NULL
)
```

Arguments

`data`	A data frame containing the full dataset.
`plot_var`	A data frame with "Variable" and "PlotTitle" columns specifying which variables to plot and their titles. If NULL, all variables in the data will be plotted with default titles.
`x_axis_from`	Character. Column name to use for x-axis grouping
`stack_value_from`	Character. Column name containing the stack categories
`title_prefix`	Optional character string to prepend to the title
`title_suffix`	Optional character string to append to the title
`y_axis_title`	Character. Y-axis label. If NULL, uses Unit if available.
`compare_by_experiment`	Logical. If TRUE, compares x-axis values within experiments
`separate_figure`	Logical. If TRUE, creates separate figures instead of panels
`unstack_plot`	Logical. If TRUE, displays components side by side instead of stacked and creates separate figures for each x_axis_from value
`output_dir`	Optional character. Directory to save the output plot
`panel_rows`	Optional numeric. Number of panel rows
`panel_cols`	Optional numeric. Number of panel columns
`color_tone`	Optional character. Base color for plot color generation
`show_total`	Logical. If `TRUE`, displays total values above stacked bars. Default is `TRUE`.
`description_as_title`	Logical. If `TRUE`, uses the "Description" column as the title instead of "Variable". Default is `TRUE`.
`legend_position`	Character. Position of the legend: "bottom" (default), "top", "left", "right", or "none"
`y_limit`	Optional numeric vector. Custom y-axis limits (e.g., c(-5, 5))
`width`	Optional numeric. Width of the output plot
`height`	Optional numeric. Height of the output plot

Value

A list of ggplot objects or a single ggplot object depending on settings

Author(s)

Pattawee Puangchit

Index